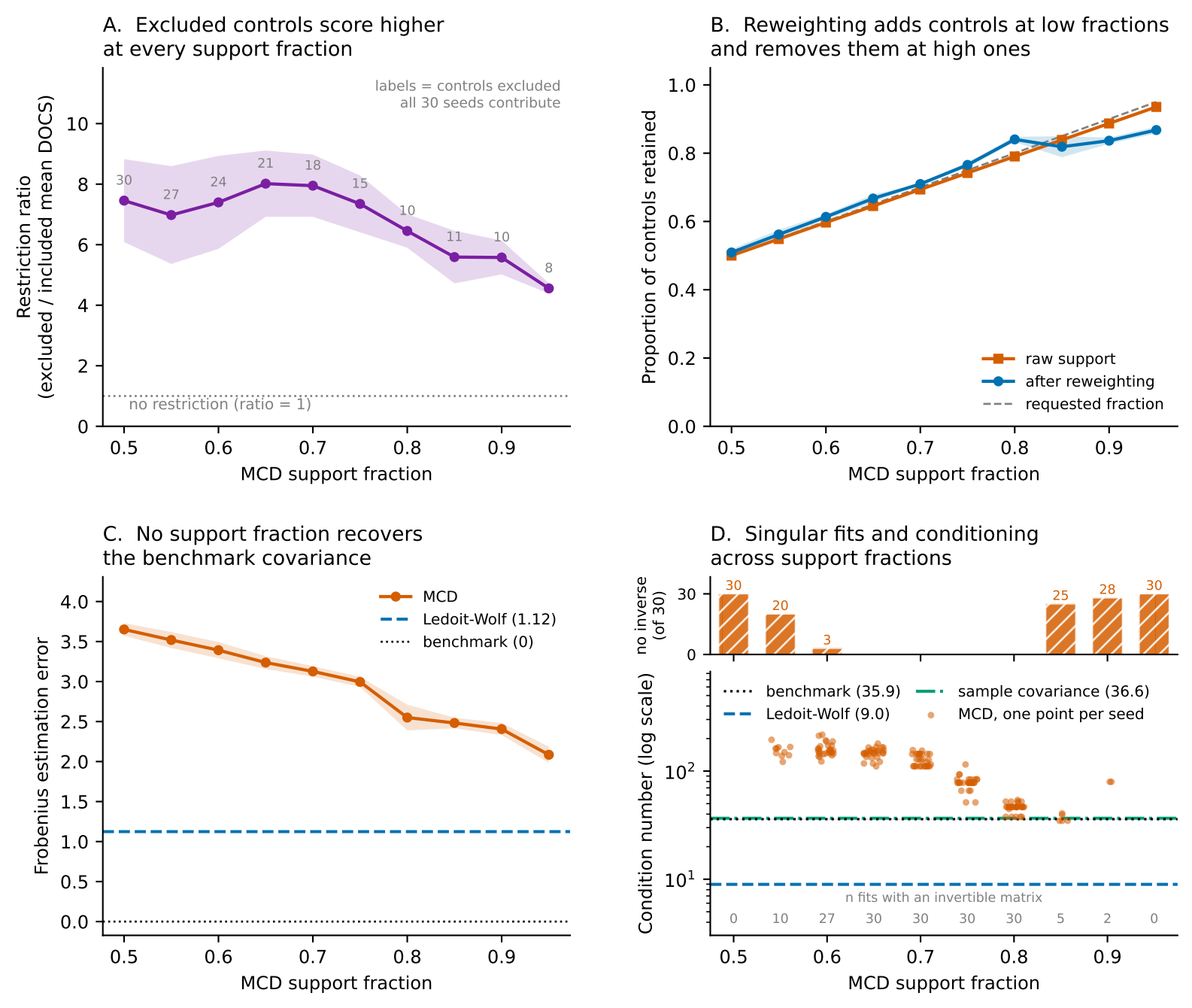
A. Excluded controls score higher at every support fraction
labels = controls excluded
all 30 seeds contribute
Restriction ratio
(excluded / included mean DOCS)
30
27
24
21
18
15
10
11
10
8
no restriction (ratio = 1)
MCD support fraction
B. Reweighting adds controls at low fractions and removes them at high ones
Proportion of controls retained
raw support
after reweighting
requested fraction
MCD support fraction
C. No support fraction recovers the benchmark covariance
Frobenius estimation error
MCD
Ledoit-Wolf (1.12)
benchmark (0)
MCD support fraction
D. Singular fits and conditioning across support fractions
no inverse (of 30)
30
20
3
25
28
30
Condition number (log scale)
benchmark (35.9)
Ledoit-Wolf (9.0)
sample covariance (36.6)
MCD, one point per seed
n fits with an invertible matrix
0
10
27
30
30
30
30
5
2
0
MCD support fraction